Secure VLAN-Based School Network Design

Domain: School / Educational Institution Network

Project Report

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Executive Summary

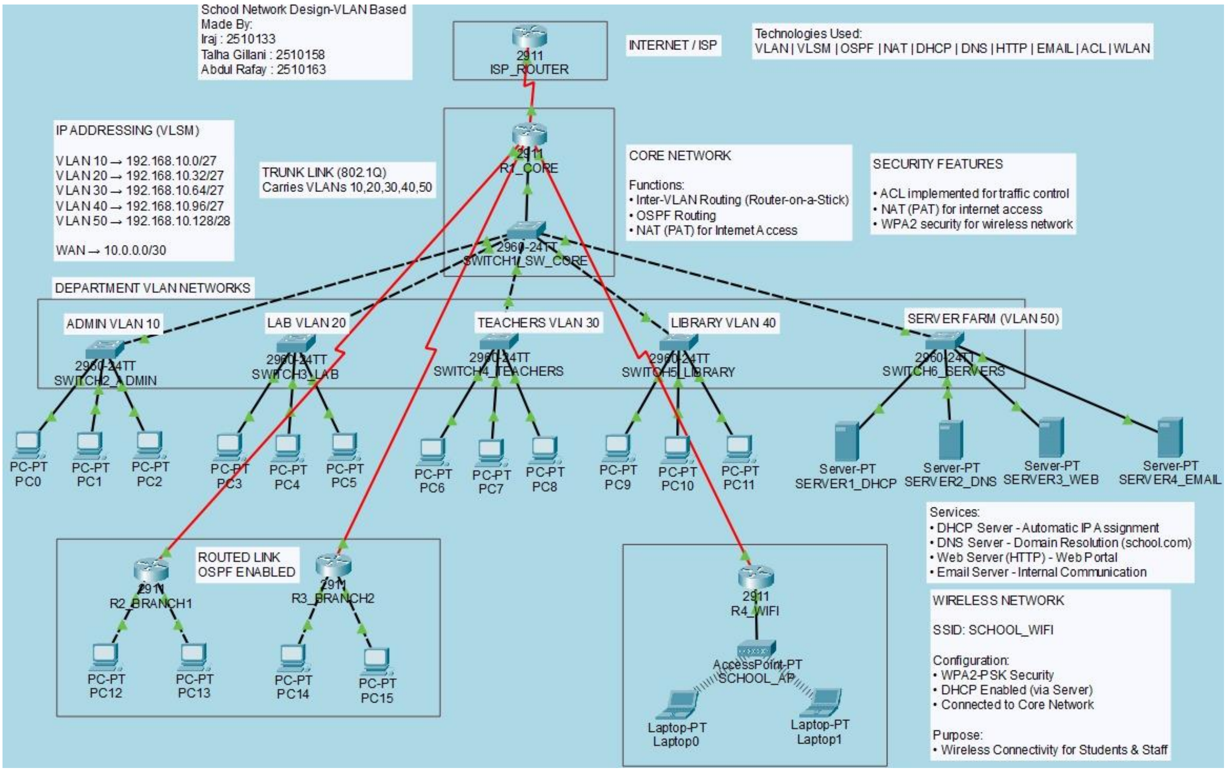
This project presents the design and implementation of a comprehensive and scalable school network using Cisco Packet Tracer. The primary objective was to develop a structured network that ensures efficient communication, resource sharing, and secure connectivity across multiple departments within an educational institution. The network design follows a hierarchical architecture consisting of core, distribution, and access layers to enhance performance, manageability, and reliability.

To achieve efficient IP utilization, Variable Length Subnet Masking (VLSM) was applied, allowing optimized allocation of address space for each department. Virtual Local Area Networks (VLANs) were implemented to logically segment the network into distinct domains, including Administration, Labs, Teachers, Library, and Server Farm, thereby improving security and reducing broadcast traffic. Inter-VLAN communication was enabled through Router-on-a-Stick configuration.

Dynamic routing using OSPF ensures efficient path selection and scalability for future expansion. Essential network services such as DHCP, DNS, HTTP, and Email were deployed to support automated configuration, domain name resolution, web access, and internal communication. Network security was reinforced through Access Control Lists (ACLs), Network Address Translation (NAT/PAT) for internet access, and WPA2-secured wireless connectivity.

The final network is fully functional, with successful connectivity verification, demonstrating a reliable and enterprise-level solution aligned with modern networking standards.

<u>**Network Topology Diagram**</u>



<u>**IP Address Plan**</u>

IP Addressing Table:

The IP addressing scheme was designed using VLSM to efficiently utilize available address space. Each VLAN was assigned a unique subnet to ensure logical segmentation and improved network performance. The server farm was allocated a smaller subnet due to fewer devices, while larger subnets were assigned to user-based VLANs. DHCP was used to dynamically assign IP addresses to end devices, reducing manual configuration and administrative overhead.

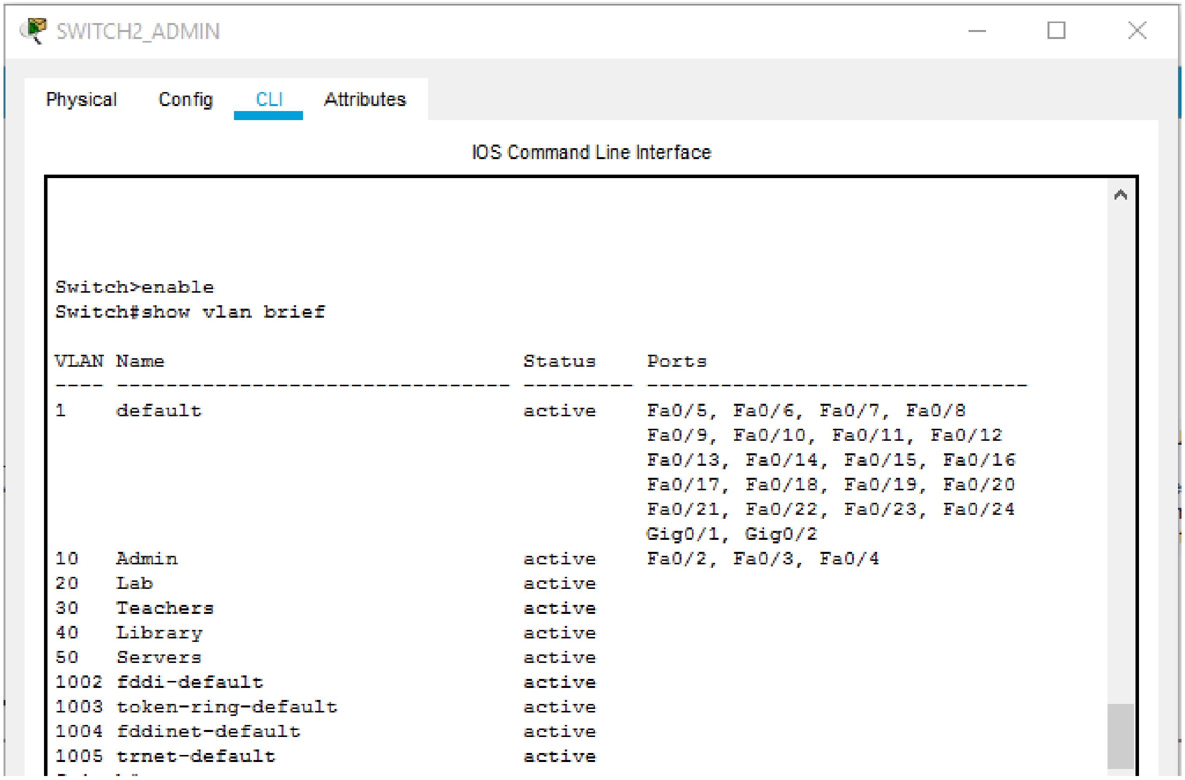
Device Name	Interface	IP Address	Subnet Mask	Default Gateway	VLAN
R1_CORE	G0/0.10	192.168.10.1	255.255.255.224	N/A	10
R1_CORE	G0/0.20	192.168.10.33	255.255.255.224	N/A	20
R1_CORE	G0/0.30	192.168.10.65	255.255.255.224	N/A	30
R1_CORE	G0/0.40	192.168.10.97	255.255.255.224	N/A	40
R1_CORE	G0/0.50	192.168.10.129	255.255.255.240	N/A	50
R1_CORE	S0/1/0	10.0.0.1	255.255.255.252	N/A	WAN
ISP_ROUTER	S0/0/0	10.0.0.2	255.255.255.252	N/A	WAN
SERVER1_DHCP	Fa0	192.168.10.130	255.255.255.240	192.168.10.129	50
SERVER2_DNS	Fa0	192.168.10.131	255.255.255.240	192.168.10.129	50
SERVER3_WEB	Fa0	192.168.10.132	255.255.255.240	192.168.10.129	50
SERVER4_EMAIL	Fa0	192.168.10.133	255.255.255.240	192.168.10.129	50
ADMIN_PC1	Fa0	DHCP	DHCP	192.168.10.1	10
LAB_PC1	Fa0	DHCP	DHCP	192.168.10.33	20
TEACHER_PC1	Fa0	DHCP	DHCP	192.168.10.65	30
LIBRARY_PC1	Fa0	DHCP	DHCP	192.168.10.97	40
LAPTOP1	Wireless	DHCP	DHCP	192.168.40.1	WLAN

Configuration Logs

This section highlights the essential configuration commands used to implement core networking functionalities including VLAN segmentation, inter-VLAN routing, dynamic routing, DHCP services, NAT, and security using ACLs.

VLAN Configuration (Switches):

```
enable

configure terminal


vlan 10

name ADMIN


vlan 20

name LAB


vlan 30

name TEACHERS


vlan 40

name LIBRARY


vlan 50

name SERVERS
```



Access Port Assignment:

interface range fa0/1-4

switchport mode access

switchport access vlan 10

spanning-tree portfast

interface range fa0/1-4

switchport mode access

switchport access vlan 20

spanning-tree portfast

interface range fa0/1-4

switchport mode access

switchport access vlan 30

spanning-tree portfast

interface range fa0/1-4

switchport mode access

switchport access vlan 40

spanning-tree portfast

Trunk Configuration:

interface fa0/1

switchport mode trunk

no shutdown

```
Switch>enable
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1
Fa0/5       on           802.1q         trunking      1
Fa0/6       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005
Fa0/5       1-1005
Fa0/6       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Fa0/5       1,10,20,30,40,50
Fa0/6       1,10,20,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,40,50
Fa0/2       1,10,20,30,40,50
Fa0/3       1,10,20,30,40,50
Fa0/4       1,10,20,30,40,50
Fa0/5       1,10,20,30,40,50
Fa0/6       1,10,20,30,40,50
```

Inter-VLAN Routing:

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.224

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.10.33 255.255.255.224

interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.10.65 255.255.255.224
interface g0/0.40
encapsulation dot1Q 40
ip address 192.168.10.97 255.255.255.224

interface g0/0.50
encapsulation dot1Q 50
ip address 192.168.10.129 255.255.255.240
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.10.33   YES manual up                    up
GigabitEthernet0/0.30  192.168.10.65   YES manual up                    up
GigabitEthernet0/0.40  192.168.10.97   YES manual up                    up
GigabitEthernet0/0.50  192.168.10.129  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
FastEthernet0/0/0      unassigned      YES unset  up                    down
FastEthernet0/0/1      unassigned      YES unset  up                    down
FastEthernet0/0/2      unassigned      YES unset  up                    down
FastEthernet0/0/3      unassigned      YES unset  up                    down
Serial0/1/0            10.0.0.1        YES manual up                    up
Serial0/1/1            10.0.0.5        YES manual up                    up
Serial0/2/0            10.0.0.9        YES manual up                    up
Serial0/2/1            10.0.0.13       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

OSPF Routing Configuration:

router ospf 1

network 192.168.10.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

```
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.21.1      0   FULL/  -        00:00:33    10.0.0.6        Serial0/1/1
192.168.31.1      0   FULL/  -        00:00:36    10.0.0.10       Serial0/2/0
192.168.40.1      0   FULL/  -        00:00:33    10.0.0.14       Serial0/2/1
```

DHCP Configuration (Server-based):

DHCP service was configured on the dedicated server by defining multiple pools for each VLAN, specifying default gateway, DNS server, and IP address ranges.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
WIFI	192.168.40.1	192.168.10.131	192.168.40.10	255.255.255.0	246	0.0.0.0	0.0.0.0
Library Pool	192.168.10.97	192.168.10.131	192.168.10.98	255.255.255....	30	0.0.0.0	0.0.0.0
Teachers Pool	192.168.10.65	192.168.10.131	192.168.10.66	255.255.255....	30	0.0.0.0	0.0.0.0
Lab Pool	192.168.10.33	192.168.10.131	192.168.10.34	255.255.255....	30	0.0.0.0	0.0.0.0
Admin Pool	192.168.10.1	192.168.10.131	192.168.10.2	255.255.255....	30	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.128	255.255.255....	255	0.0.0.0	0.0.0.0

NAT (PAT) Configuration:

access-list 1 permit 192.168.10.0 0.0.0.255

interface g0/0

ip nat inside

interface serial0/0/0

ip nat outside

ip nat inside source list 1 interface serial0/0/0 overload

ip route 0.0.0.0 0.0.0.0 10.0.0.2

ACL Configuration (Security):

access-list 100 deny ip 192.168.10.0 0.0.0.31 192.168.10.64 0.0.0.31

access-list 100 permit ip any any

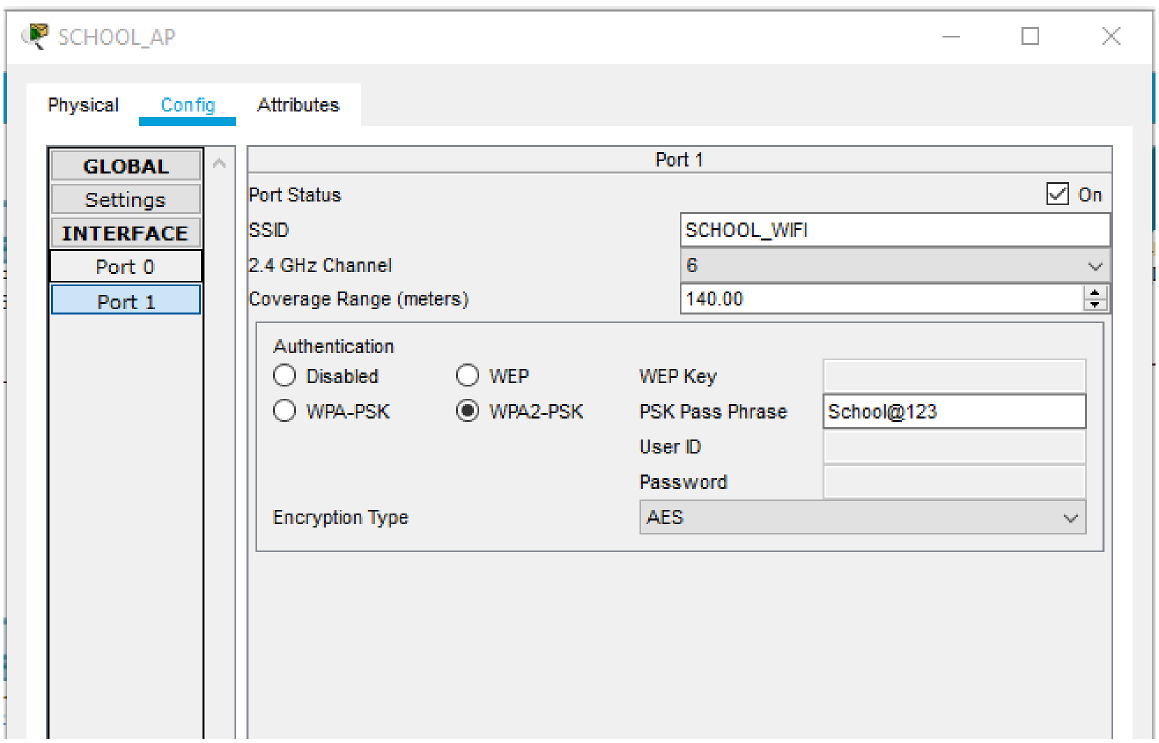
interface g0/0

ip access-group 100 in

```
Router#show access-lists
Standard IP access list 1
    10 permit 192.168.10.0 0.0.0.255
Extended IP access list 101
    10 deny ip 192.168.10.32 0.0.0.31 192.168.10.0 0.0.0.31 (8 match(es))
    20 permit ip any any (61 match(es))
```

Wireless Configuration:

Wireless network configured using an Access Point with SSID "SCHOOL_WIFI". WPA2-PSK security was enabled, and clients obtained IP addresses dynamically via DHCP.



The configurations were tested and verified using show commands and connectivity tests to ensure correct implementation and functionality of all networking components.

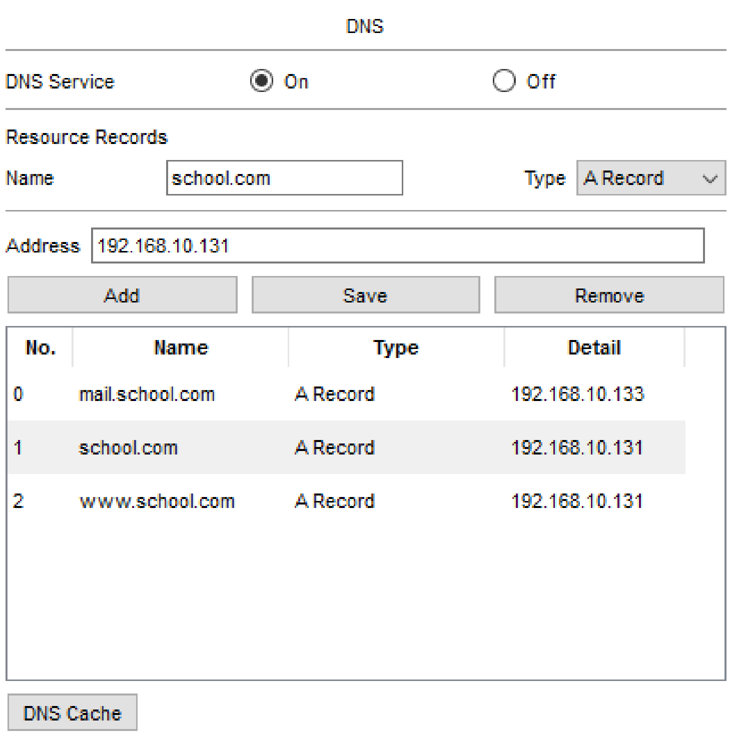
Service Configuration

This section demonstrates the configuration and successful operation of essential network services including DHCP, DNS, HTTP, and Email. Screenshots are provided as evidence of proper service setup and functionality.

DHCP Configuration Screenshot:

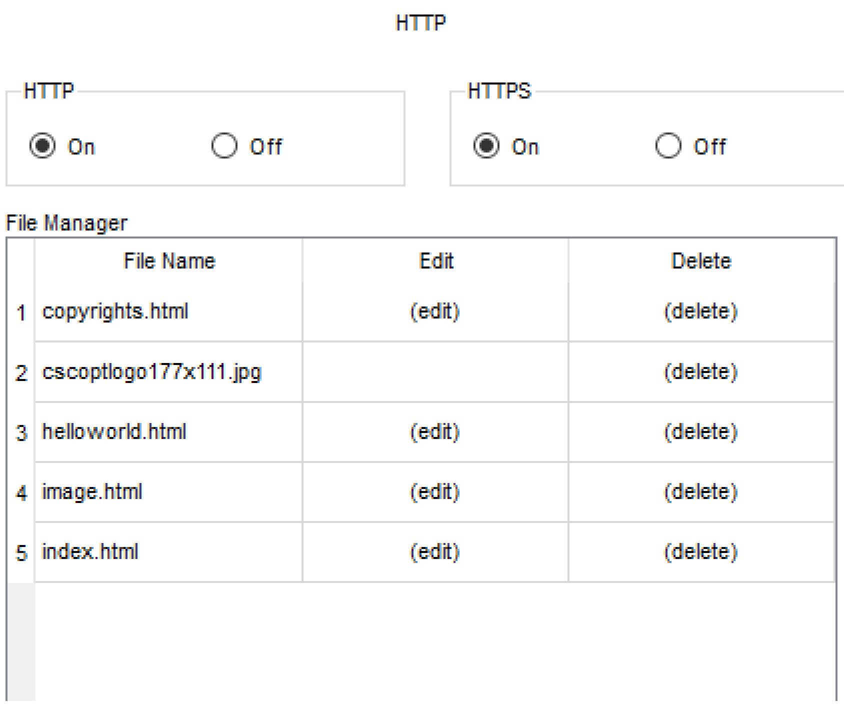
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
WIFI	192.168.40.1	192.168.10.131	192.168.40.10	255.255.255.0	246	0.0.0.0	0.0.0.0
Library Pool	192.168.10.97	192.168.10.131	192.168.10.98	255.255.255.224	30	0.0.0.0	0.0.0.0
Teachers Pool	192.168.10.65	192.168.10.131	192.168.10.66	255.255.255.224	30	0.0.0.0	0.0.0.0
Lab Pool	192.168.10.33	192.168.10.131	192.168.10.34	255.255.255.224	30	0.0.0.0	0.0.0.0
Admin Pool	192.168.10.1	192.168.10.131	192.168.10.2	255.255.255.224	30	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.128	255.255.255.240	255	0.0.0.0	0.0.0.0

DNS Configuration Screenshot:

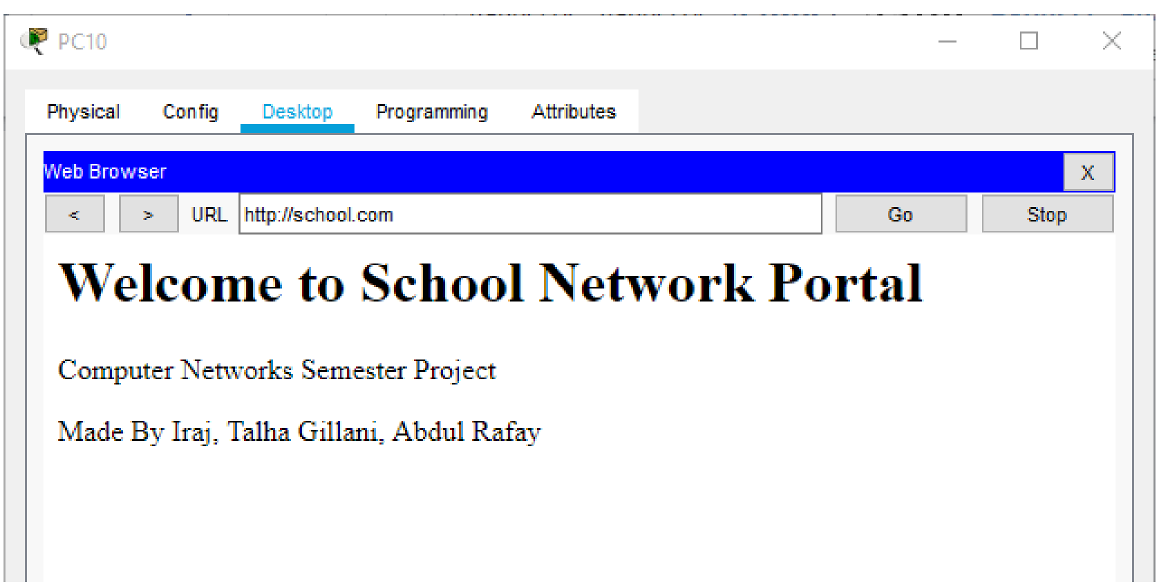


HTTP (Web Server) Screenshot:

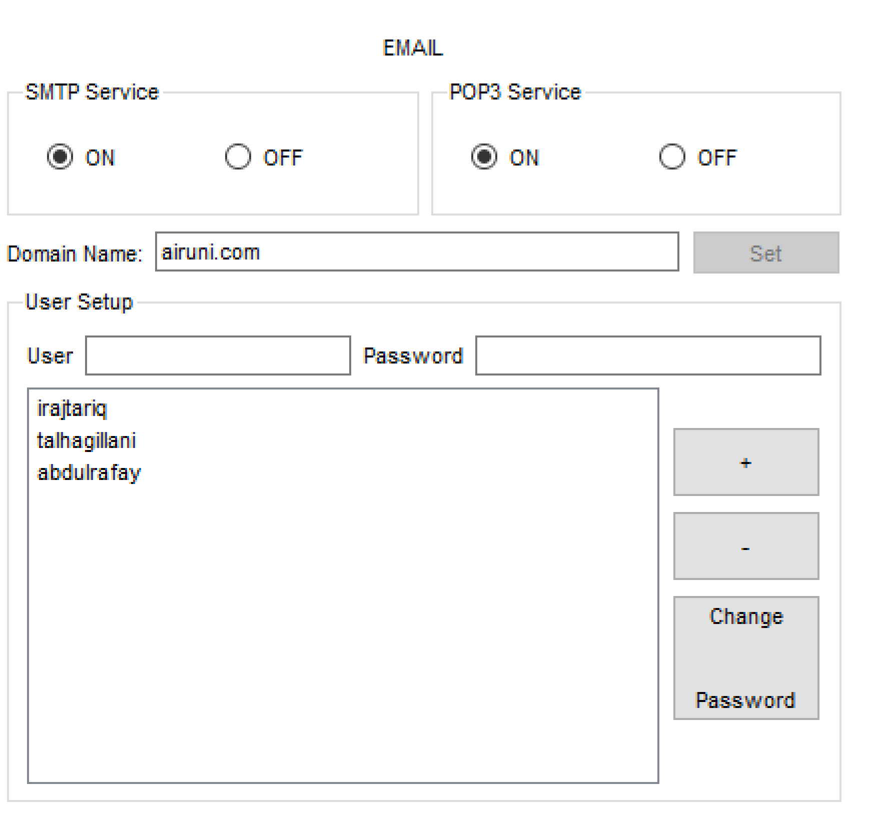
Server Side:



Client Side:



EMAIL Service Screenshot:



Security Policy

To enhance network security and control inter-departmental communication, Access Control Lists (ACLs) were implemented within the network. The purpose of these ACLs is to restrict unauthorized access between VLANs while still allowing necessary services such as DNS, HTTP, DHCP, and Email to function correctly.

The policy focuses on limiting direct communication between user VLANs (e.g., Admin, Lab, Teachers, Library) while permitting access to shared resources located in the Server Farm (VLAN 50).

Rule No.	Action	Source Network	Destination Network	Purpose
1	DENY	VLAN 10 (Admin)	VLAN 30 (Teachers)	Prevent unauthorized inter-department access
2	DENY	VLAN 20 (Lab)	VLAN 10 (Admin)	Restrict student access to admin systems
3	PERMIT	Any VLAN	VLAN 50 (Servers)	Allow access to shared services (DNS, Web, Email)
4	PERMIT	Any	Any	Allow all other legitimate traffic

ACL Configuration (CLI):

```
Router#show access-lists
Standard IP access list 1
    10 permit 192.168.10.0 0.0.0.255
Extended IP access list 101
    10 deny ip 192.168.10.32 0.0.0.31 192.168.10.0 0.0.0.31 (8 match(es))
    20 permit ip any any (61 match(es))
```

Justification:

The implemented ACL policy ensures that sensitive departments such as Administration are protected from unauthorized access by other VLANs, particularly student-based networks such as the Lab VLAN. This minimizes the risk of data breaches and unauthorized system usage.

At the same time, access to the Server Farm is permitted from all VLANs to ensure uninterrupted availability of critical services such as DHCP, DNS, HTTP, and Email. This balance between restriction and accessibility ensures both security and usability within the network.

The final rule permitting all remaining traffic ensures that normal network operations are not disrupted, maintaining overall network functionality while enforcing targeted restrictions.

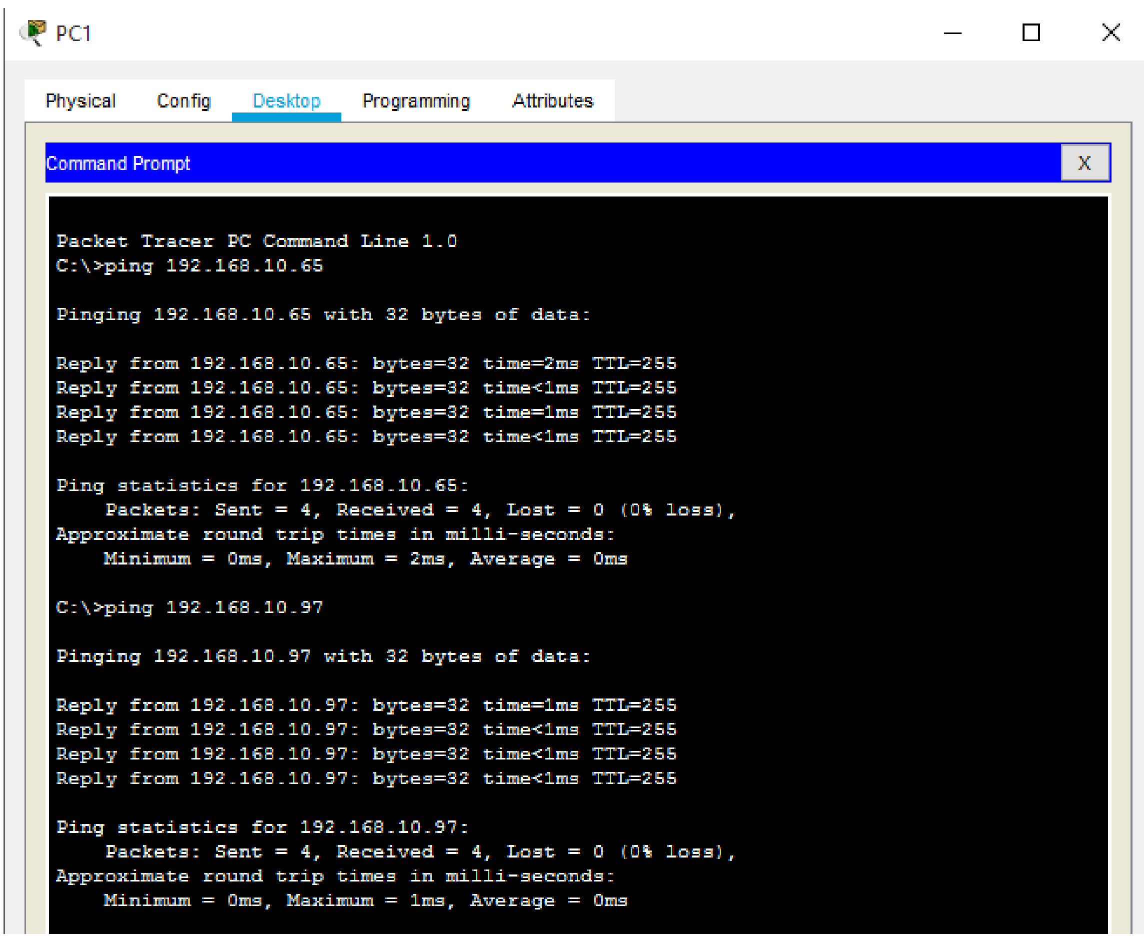
Verification Evidence

This section presents verification results demonstrating the correct functionality of the implemented network. Connectivity tests, routing verification, and service validation were performed using ping, traceroute, and show commands.

Ping Tests:

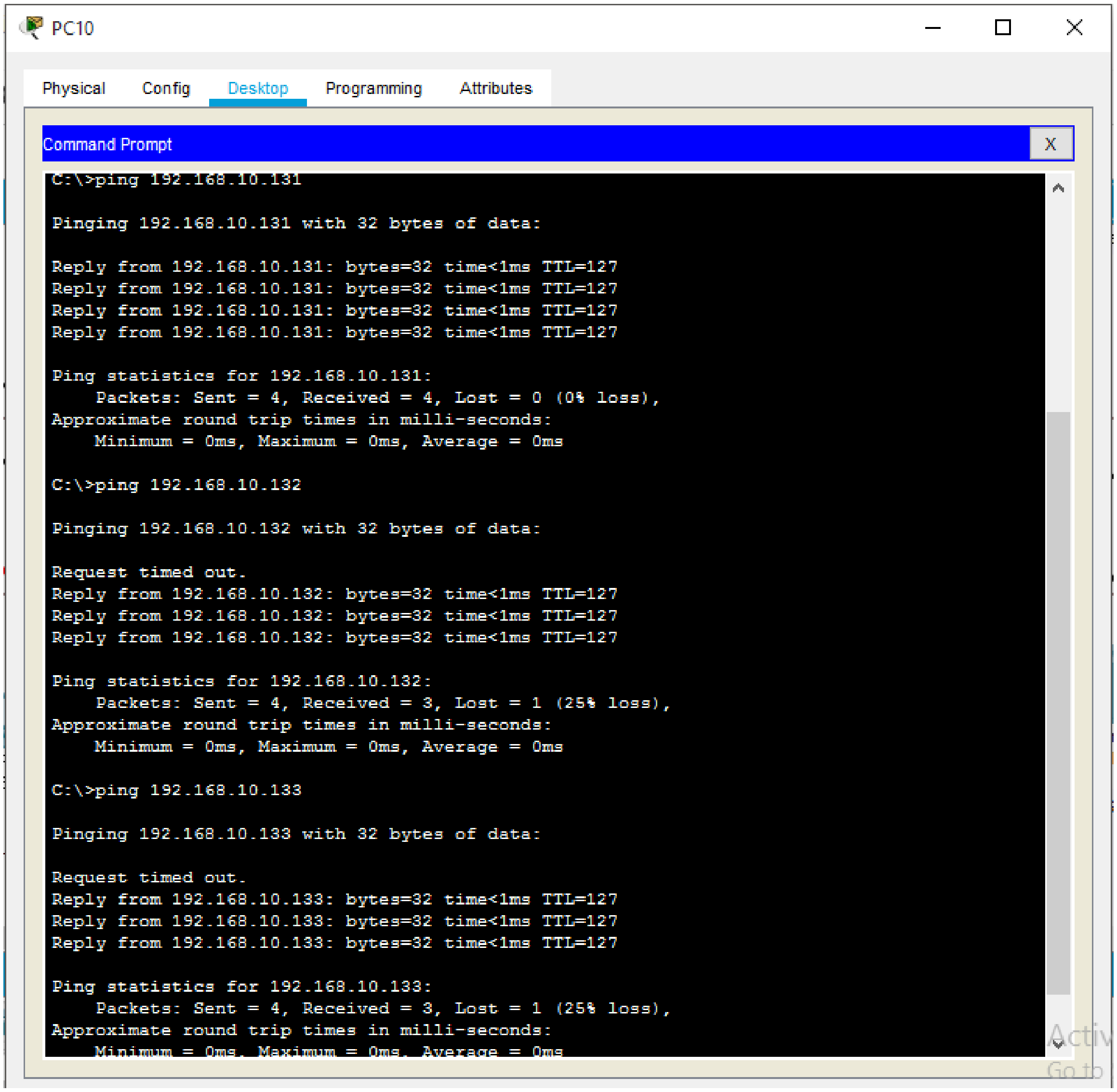
Inter-VLAN Connectivity:

Test from Admin PC:



Server Connectivity:

Pinging Servers from Any PC:



Internet (NAT) Test:

Successful communication with ISP network through NAT.

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=6ms TTL=254

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 6ms, Average = 2ms
```

Traceroute Outputs:

```
C:\>tracert 192.168.10.131

Tracing route to 192.168.10.131 over a maximum of 30 hops:

  1   1 ms      0 ms      0 ms      192.168.10.65
  2   0 ms      0 ms      0 ms      192.168.10.131

Trace complete.

C:\>tracert 192.168.10.132

Tracing route to 192.168.10.132 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.65
  2   0 ms      0 ms      0 ms      192.168.10.132

Trace complete.
```

Routing Verification:

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.1/32 is directly connected, Serial0/1/0
C       10.0.0.4/30 is directly connected, Serial0/1/1
L       10.0.0.5/32 is directly connected, Serial0/1/1
C       10.0.0.8/30 is directly connected, Serial0/2/0
L       10.0.0.9/32 is directly connected, Serial0/2/0
C       10.0.0.12/30 is directly connected, Serial0/2/1
L       10.0.0.13/32 is directly connected, Serial0/2/1
     192.168.10.0/24 is variably subnetted, 10 subnets, 3 masks
C       192.168.10.0/27 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.10.32/27 is directly connected, GigabitEthernet0/0.20
L       192.168.10.33/32 is directly connected, GigabitEthernet0/0.20
C       192.168.10.64/27 is directly connected, GigabitEthernet0/0.30
L       192.168.10.65/32 is directly connected, GigabitEthernet0/0.30
C       192.168.10.96/27 is directly connected, GigabitEthernet0/0.40
L       192.168.10.97/32 is directly connected, GigabitEthernet0/0.40
C       192.168.10.128/28 is directly connected, GigabitEthernet0/0.50
L       192.168.10.129/32 is directly connected, GigabitEthernet0/0.50
O    192.168.40.0/24 [110/65] via 10.0.0.14, 00:13:08, Serial0/2/1
S*   0.0.0.0/0 [1/0] via 10.0.0.2
```

Nat Verification:

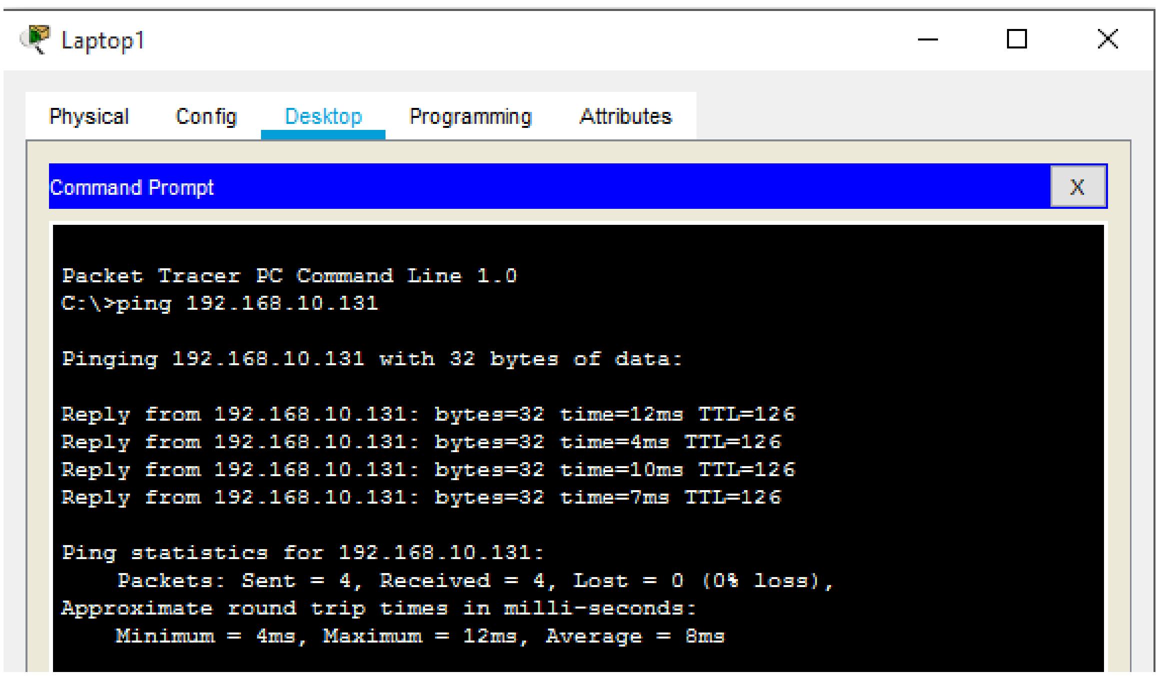
```
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/1/0
Inside Interfaces: GigabitEthernet0/0
Hits: 0  Misses: 28
Expired translations: 0
Dynamic mappings:
```

VLAN Verification:

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   Admin                            active
20   Lab                              active
30   Teachers                         active    Fa0/2, Fa0/3, Fa0/4
40   Library                          active
50   Servers                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Wireless Verification:

Ping from Laptop:



All verification tests were successfully completed, confirming full connectivity between VLANs, proper operation of network services, successful routing via OSPF, and internet access through NAT. The results demonstrate that the network is fully functional and meets all project requirements.

Challenges & Solutions

During the implementation of the network topology, several technical challenges were encountered related to VLAN configuration, routing, DHCP functionality, service accessibility, and WAN connectivity. Each issue was systematically diagnosed and resolved using troubleshooting techniques and verification commands. The following section outlines the major problems faced and the corresponding solutions applied.

1. Native VLAN Mismatch & Trunk Issues:

Problem:

While configuring trunk links between switches and the core router, native VLAN mismatch errors occurred. This caused instability in communication and prevented proper VLAN traffic flow.

Solution:

The issue was resolved by removing inconsistent native VLAN configurations and standardizing trunk links across all switches. Default VLAN settings were restored to ensure uniformity, after which trunk links began functioning correctly.

2. Incorrect Interface Configuration:

Problem:

Errors were encountered when configuring router interfaces, where certain interfaces appeared unavailable or invalid during configuration.

Solution:

This was resolved by verifying the correct interface names using 'show ip interface brief' and ensuring that the correct physical interfaces were used. Misconfigured interfaces were corrected, allowing proper routing configuration.

3. DHCP Failure (APIPA Addressing):

Problem:

Several end devices failed to obtain IP addresses dynamically and instead assigned APIPA addresses (169.254.x.x), indicating DHCP failure.

Solution:

The root cause was identified as missing DHCP relay configuration. This was resolved by applying 'ip helper-address' on router subinterfaces, enabling DHCP requests to reach the DHCP server across VLANs.

4. VLAN & Trunk Propagation Issues:

Problem:

Certain switches did not show active trunk links, preventing VLAN traffic from reaching the core network and isolating devices.

Solution:

The issue was resolved by correctly identifying uplink ports using 'show cdp neighbors' and configuring them explicitly as trunk ports. VLANs were also created on all switches to ensure proper propagation.

5. Server Connectivity Failure:

Problem:

End devices were unable to communicate with servers despite successful router connectivity, indicating VLAN or gateway misconfiguration**.**

Solution:

The issue was resolved by verifying VLAN assignments on switch ports and ensuring correct default gateway configuration for each VLAN. Once corrected, communication with servers was restored.

6. DNS Resolution Failure:

Problem:

The domain 'school.com' failed to resolve on client devices, preventing access to the web server.

Solution:

The DNS server configuration was corrected by properly mapping the domain name to the correct web server IP address. Once updated, domain resolution functioned successfully.

7. Wireless Connectivity Issues:

Problem:

Wireless devices failed to connect due to missing wireless interface modules and inability to obtain IP addresses.

Solution:

Wireless NIC modules were installed on laptops, and DHCP relay was configured on the wireless router interface. This enabled successful connectivity and IP assignment.

8. WAN / NAT Failure

Problem:

Internal devices were unable to reach the ISP network, and NAT functionality failed.

Solution:

The issue was identified as a missing IP address configuration on the ISP router's serial interface. After assigning the correct IP and verifying the link status, WAN connectivity and NAT translation began working correctly.

The challenges encountered during the implementation process significantly enhanced understanding of real-world networking issues. Through systematic troubleshooting, use of diagnostic commands, and iterative testing, all issues were successfully resolved, resulting in a fully functional and stable network.

Conclusion

The successful completion of this project demonstrated the design and implementation of a fully functional enterprise-style school network using Cisco Packet Tracer. The network was developed to provide efficient, secure, and scalable communication between multiple departments within the institution while meeting all specified project requirements.

Throughout the project, essential networking technologies and concepts were successfully implemented, including VLSM-based IP addressing, VLAN segmentation, inter-VLAN routing using Router-on-a-Stick, OSPF dynamic routing, DHCP for automatic IP allocation, DNS for domain name resolution, HTTP web services, internal Email communication, NAT/PAT for internet connectivity, ACL-based traffic filtering, and WPA2-secured wireless networking. The final topology provided reliable communication between all network zones and ensured centralized access to shared services hosted within the server farm.

The project also emphasized practical troubleshooting and network verification techniques. Multiple connectivity tests, routing verifications, NAT translations, service validations, and wireless communication checks confirmed that the network operated correctly and efficiently. The implementation process provided valuable hands-on experience in configuring and managing a complex multi-service network environment similar to real-world organizational infrastructures.

In a real deployment environment, additional enterprise-level enhancements could further improve the network’s performance, scalability, and security. These enhancements may include redundant routers and links for high availability, dedicated hardware firewalls, intrusion detection and prevention systems (IDS/IPS), centralized network monitoring platforms, VPN support for remote access, advanced authentication systems, cloud-managed infrastructure, and backup disaster recovery solutions. Furthermore, implementing Layer 3 switches, load balancing, and advanced wireless controllers would improve network efficiency and support future expansion.

Overall, this project successfully achieved its objectives while demonstrating a comprehensive understanding of modern computer networking principles, network security practices, service integration, and enterprise network design methodologies.